

NexaSat Digital Wallet: AI-Driven Customer Lifetime Value Prediction for Customer Analytics and Personalised Marketing

Business Overview / Problem

NexaSat Digital Wallet is a rapidly growing fintech company that provides secure digital payment services, peer-to-peer money transfers, bill payments, merchant payments, savings solutions, and other financial services through its mobile wallet platform. As the company expands its customer base and introduces new financial products, it faces the challenge of understanding which customers generate the greatest long-term value and how to maximise customer profitability through personalised engagement.

The digital wallet serves a diverse customer population, ranging from occasional users who perform only basic transactions to highly engaged customers who frequently use multiple financial services. Applying the same marketing strategy across all customers has resulted in inefficient customer acquisition, retention, and engagement efforts. To improve return on marketing investment, NexaSat requires a data-driven approach to identify high-value customers, encourage long-term loyalty, and allocate resources more effectively.

In an increasingly competitive fintech landscape, where customers can easily switch between digital payment providers, retaining existing customers has become just as important as acquiring new ones. NexaSat recognises that increasing Customer Lifetime Value (CLV) is essential for sustainable revenue growth. The company has identified significant opportunities to increase Average Revenue Per User (ARPU) by promoting additional financial services such as digital savings accounts, micro-loans, insurance products, cashback programmes, investment services, premium wallet subscriptions, and merchant payment solutions. However, these opportunities depend on accurately identifying customers who are most likely to adopt these products.

Although NexaSat collects large volumes of transactional, behavioural, demographic, and engagement data, it currently lacks a systematic framework for transforming this information into actionable customer insights. Marketing campaigns are often broad and untargeted, resulting in lower conversion rates, unnecessary promotional expenditure, and missed revenue opportunities. Furthermore, customer retention initiatives are reactive rather than proactive, making it difficult to identify customers who are at risk of churn before they disengage from the platform.

The primary business challenge is therefore to develop an AI-driven Customer Lifetime Value prediction framework that accurately estimates the future value of each customer, segments customers based on predicted lifetime value, and identifies those with the highest potential for up-selling and cross-selling. Such a solution would enable NexaSat to deliver personalised marketing campaigns, improve customer retention, optimise promotional spending, and strengthen long-term customer relationships.

By leveraging machine learning and advanced customer analytics, NexaSat Digital Wallet can transition from reactive customer management to a proactive, intelligence-driven strategy.



Predictive Customer Lifetime Value modelling enables the business to prioritise high-value customers, identify emerging high-potential users, predict customer churn, and recommend personalised financial products based on individual transaction behaviour and spending patterns. Ultimately, this supports increased customer satisfaction, higher product adoption, improved customer retention, and sustainable revenue growth while providing a competitive advantage in the rapidly evolving digital payments ecosystem.

Certainly. Since your project is **Customer Lifetime Value Prediction** rather than **Customer Lifetime Value Segmentation**, the rationale and aim should focus on **predictive modelling**, **customer analytics**, and **business value** without mentioning segmentation.

Rationale for the Project

Customer Lifetime Value (CLV) prediction is a key customer analytics technique that uses historical customer behaviour and machine learning to estimate the future value each customer is expected to generate throughout their relationship with a business. By accurately predicting customer lifetime value, organisations can make informed decisions regarding customer retention, personalised marketing, resource allocation, and revenue optimisation.

For **NexaSat Digital Wallet**, understanding customer lifetime value is essential in an increasingly competitive fintech environment where customers have access to multiple digital payment platforms and financial service providers. Although NexaSat collects extensive customer transaction and behavioural data, it currently lacks an intelligent framework to accurately predict which customers are likely to generate the greatest long-term value and which customers require targeted engagement to improve retention.

By implementing an AI-driven Customer Lifetime Value prediction model, NexaSat can estimate the future economic value of individual customers using historical transaction patterns, digital wallet usage, and customer behaviour. These predictive insights enable the company to deliver personalised marketing campaigns, improve customer retention, identify customers at risk of churn, and uncover opportunities for up-selling and cross-selling financial products such as digital savings accounts, micro-loans, insurance services, investment products, premium wallet subscriptions, and merchant payment solutions.

Ultimately, AI-driven Customer Lifetime Value prediction enables NexaSat to allocate marketing resources more effectively, improve customer engagement, maximise customer profitability, and strengthen long-term customer relationships. This data-driven approach supports sustainable business growth while providing a competitive advantage within the rapidly evolving digital payments and fintech industry.

Aim of the Project

The primary aim of this project is to develop an **AI-driven Customer Lifetime Value (CLV) prediction model** that accurately estimates the future value of customers using transactional, behavioural, and demographic data from the NexaSat Digital Wallet platform.



Using machine learning techniques, the project seeks to identify the key factors that influence customer lifetime value and generate accurate predictions that support strategic business decision-making. These insights will enable NexaSat to implement personalised marketing campaigns, improve customer retention, optimise marketing investment, and identify opportunities for up-selling and cross-selling financial products.

Specifically, the project aims to:

- Develop an end-to-end machine learning pipeline for Customer Lifetime Value prediction.
- Identify the key customer characteristics and behavioural patterns that influence long-term customer value.
- Build and evaluate predictive machine learning models capable of estimating future Customer Lifetime Value.
- Support personalised marketing and product recommendations based on predicted customer value.
- Enable proactive customer retention by identifying customers whose future lifetime value may decline.
- Provide actionable business insights through interactive dashboards and data visualisation to support strategic decision-making.

Ultimately, the project aims to help NexaSat Digital Wallet maximise customer profitability, improve customer retention, optimise marketing expenditure, and drive sustainable revenue growth through AI-powered customer analytics.

Project Scope

This project analyses **NexaSat Digital Wallet customer data from July 2023 onwards** to develop and evaluate an AI-driven Customer Lifetime Value prediction model. The dataset includes customer demographics, transaction history, digital wallet usage, payment behaviour, account activity, and customer engagement metrics. These data will be used to engineer predictive features, train and evaluate machine learning models, and generate actionable insights that support data-driven decision-making. The resulting solution demonstrates how customer analytics and artificial intelligence can improve customer relationship management and long-term business performance within the fintech industry.

Tech Stack

The **NexaSat Digital Wallet: AI-Driven Customer Lifetime Value Prediction for Customer Analytics** project was developed using a modern data science, machine learning, and business intelligence technology stack to support the complete analytics lifecycle, from data extraction and preprocessing to predictive modelling, model interpretation, deployment, and business reporting.



Programming Languages

Python was used as the primary programming language for data preprocessing, exploratory data analysis, feature engineering, machine learning model development, model evaluation, and predictive analytics.

SQL was used for querying, extracting, and transforming customer transaction data, enabling efficient customer behaviour analysis and feature generation.

Data Processing and Analysis

The project utilised **Pandas** and **NumPy** for data cleaning, transformation, feature engineering, and numerical analysis. These libraries supported the preparation of customer demographic, transactional, and behavioural data required for machine learning development.

Data Visualisation

Plotly is used for exploratory data analysis and visualisation of customer behaviour patterns, transaction trends, feature relationships, and model evaluation results.

Machine Learning

Machine learning models were developed using Scikit-learn, Tensorflow and advanced gradient boosting frameworks to predict Customer Lifetime Value (CLV). The modelling approach incorporated a diverse range of regression techniques, including linear regression, decision trees, ensemble learning, kernel-based learning, neural networks, and advanced gradient boosting algorithms. This allowed the analysis to capture both linear and complex non-linear relationships between customer demographics, behavioural patterns, transaction characteristics, engagement, and future customer value.

The following regression models were evaluated:

- **Linear Regression** – A baseline statistical learning algorithm used to establish a benchmark relationship between customer characteristics and Customer Lifetime Value. Its simplicity and interpretability provide a reference point for assessing the performance of more complex machine learning models.
- **Decision Tree Regressor** – A non-linear tree-based algorithm that recursively partitions the dataset based on customer characteristics to generate LTV predictions. The model can capture non-linear relationships and interactions between customer features without requiring assumptions of linearity.
- **Random Forest Regressor** – A bagging ensemble algorithm that combines multiple decision trees to improve predictive accuracy, reduce overfitting, and capture complex relationships within customer behaviour and transaction patterns.
- **Support Vector Regression (SVR)** – A kernel-based regression algorithm capable of modelling both linear and non-linear relationships. SVR was included to assess whether a margin-based learning approach could effectively predict customer lifetime value from behavioural and transactional characteristics.
- **MLP Regressor (Multi-Layer Perceptron)** – A neural-network-based regression model consisting of interconnected layers of neurons that learn complex non-linear patterns

from the customer data. MLP was used to assess the ability of neural networks to capture higher-order relationships and interactions influencing customer lifetime value.

- XGBoost Regressor – An advanced gradient boosting algorithm that sequentially builds decision trees to minimise prediction errors. XGBoost was selected for its strong predictive performance, regularisation capabilities, computational efficiency, and ability to model complex non-linear relationships within structured customer data.
- CatBoost Regressor – An advanced gradient boosting algorithm designed to provide strong predictive performance while efficiently handling categorical variables. CatBoost was included because the NexaSat dataset contains customer characteristics that may benefit from effective categorical-feature processing and its ability to model complex feature interactions.

Overall, the combination of these models provides a comprehensive comparison of different machine learning paradigms, enabling the identification of the most accurate and robust approach for predicting Customer Lifetime Value.

Model performance was evaluated using regression evaluation metrics including **Mean Absolute Error (MAE)**, **Root Mean Squared Error (RMSE)**, **Mean Absolute Percentage Error (MAPE)**, and **R² Score** to identify the most accurate model for predicting Customer Lifetime Value.

Model Explainability

SHAP (SHapley Additive Explanations) was used to improve model transparency by analysing feature contributions and explaining how customer characteristics and behaviours influence predicted Customer Lifetime Value. This supports business stakeholders in understanding the key drivers behind customer value predictions.

Model Deployment

The final Customer Lifetime Value prediction model was deployed using **Amazon Web Service (AWS)** enabling the development of an interactive machine learning application. The deployment allows users to input customer characteristics and behavioural attributes, generate CLV predictions, and explore model outputs through an intuitive web interface.

AWS provided a lightweight framework for transforming the developed machine learning model into an accessible analytical application suitable for customer analytics and business decision support.

Business Intelligence and Dashboard Development

Microsoft Power BI was used to develop interactive dashboards and business intelligence reports. The dashboard provides stakeholders with actionable insights into customer value predictions, customer behaviour patterns, revenue opportunities, and model performance. Power BI enabled:

- Interactive customer analytics dashboards
- Customer value monitoring
- Transaction and behavioural trend analysis

- Key performance indicator (KPI) reporting
- Data-driven decision-making for customer engagement strategies

Database Management

PostgreSQL was used for storing and managing customer transaction and behavioural datasets. SQL queries were used to extract, aggregate, and transform data before integration into the machine learning pipeline.

Development Environment

Jupyter Notebook was used for exploratory data analysis, experimentation, statistical analysis, and machine learning model development.

Visual Studio Code (VS Code) was used for software development, project organisation, testing, and implementation of the analytics workflow.

Version Control

Git and **GitHub** were used for source code management, version control, project documentation, and maintaining a structured development workflow.

Artificial Intelligence Development Tools

AI-assisted development tools were incorporated to improve productivity, analytical capability, code quality, and project documentation:

- **ChatGPT (OpenAI):** Used for code generation, debugging, technical explanations, documentation support, and machine learning workflow optimisation.
- **Claude AI:** Used for code review, analytical reasoning, solution refinement, and improving technical documentation.
- **GitHub Copilot:** Used for AI-assisted coding, code completion, and accelerating software development.
- **Microsoft Copilot:** Used to support business intelligence workflows, reporting, data storytelling, and productivity within the Microsoft ecosystem.
- **Google Gemini:** Used for AI-assisted research, technical exploration, analytical problem-solving, and exploring alternative approaches to machine learning solutions.

Together, these technologies enabled the development of a robust AI-driven Customer Lifetime Value prediction solution that combines machine learning, customer analytics, model deployment, and business intelligence to support data-driven decision-making within the fintech industry.